

Aryak Bhattacharya

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I am a MS in Artificial Intelligence at DePaul University with experience in Computer Vision, NLP, Reinforcement Learning, and Agentic AI systems. Skilled in PyTorch, TensorFlow, Hugging Face, FAISS, and OpenAI APIs. Seeking AI/ML Engineer, Computer Vision, or Generative AI opportunities.

EDUCATION

DePaul University – Chicago	2024 – Present (Expected 2026)
Master of Science in Artificial Intelligence	
Techno Main Salt Lake - Kolkata	2021 – 2024
Bachelor of Technology in Computer Science and Engineering	

PROJECTS

Building Footprint Segmentation (Satellite Imagery)

Tech Stack: Python, PyTorch, U-Net, DeepLabV3+, SegFormer, SpaceNet Dataset, OpenCV, Numpy

- Developing a deep learning pipeline for building footprint segmentation using U-Net, DeepLabV3+, and SegFormer
- Evaluating performance using IoU, Dice coefficient, and building count error for both pixel-level and object-level analysis

Agentic AI RAG System

Tech Stack: Python, FAISS, SentenceTransformers, Hugging Face Transformers, RAG, Vector Databases, LLMs, OpenAI API

- Built retrieval-augmented system using FAISS and SentenceTransformers embeddings
- Designed reasoning and guardrail tools for grounded responses

Desktop AI Assistant “Luna”

Tech Stack: Python, PyQt5, Speech Recognition, Google APIs, LLM Integration, Automation, SQLite, OpenAI APIs

- Developing a desktop assistant integrating voice commands, system control, and LLM-based interaction.
- Integrated Google APIs (Calendar, Drive) for task automation and productivity workflows.
- Built GUI with chat interface, auto-scroll, and real-time interaction; extending to full voice-based control

NLP to SQL with Error Correction

Tech Stack: Python, PyTorch, GRU, T5, Hugging Face Transformers, Text-to-SQL, NLP

- Built GRU-based token correction + T5 SQL generation pipeline
- Improved robustness to noisy user queries and schema mismatches
- Achieved **98%** training and **86%** test accuracy

Person Re-Identification Using Feature Fusion

Tech Stack: Python, PyTorch, ResNet-18, OpenCV, HOG, LBP, Scikit-learn

- Combined ResNet-18 embeddings with HOG, LBP, and Fourier features
- Improved Rank-1 accuracy from **4.4%** to **57.5%**

Shadow & Reflection Removal using GANs

Tech Stack: Python, PyTorch, Pix2Pix GAN, OpenCV, Sobel Edge Detection, Image Processing, ISTD Dataset, SIRR Dataset

- Built hybrid pipeline combining classical masks (Sobel, thresholding) with Pix2Pix GAN
- Designed 5-channel input (RGB + mask + edges) to guide generation

Dreamer-based World Model (Reinforcement Learning)

Tech Stack: Python, PyTorch, Dreamer, RSSM, Actor-Critic, Reinforcement Learning, Gymnasium

- Implemented RSSM-based world model with actor-critic architecture in PyTorch
- Trained RSSM-based world model for autonomous driving simulation, reducing prediction error to **16.23** MSE.

SKILLS

- **Programming:** Python, Java, C++, SQL
- **ML/DL:** PyTorch, TensorFlow, Scikit-learn, Hugging Face
- **Computer Vision:** OpenCV, U-Net, DeepLabV3+, SegFormer, ResNet
- **NLP & LLMs:** Transformers, T5, RAG, FAISS, SentenceTransformers
- **Databases & Tools:** PostgreSQL, Git, Linux, Google Colab

EXTRA-CURRICULAR ACTIVITIES

Treasurer – Technical Club

- **Managed** budgets and sponsorships for college events.